
Measurement Protocol

Measurement Object	LTE Band 25_20M_QPSK_1RB_0Offset_EVS NB 24.4kbps_Ch26340_2N
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Project	612311
Report Generation Date	26/03/13 18:41
Responsible Person	AC01-KS

Status Overview

SMD	Status	Single Value Description	Single Value	Object
Overall Receive Delay NB	Done	Delay (Cross) [ms]	147.4	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_2N
5.1b Receive Volume Control Performance 2N NB	Ok	Speech Level [dB[SPL]]	80.40	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_2N
5.2 RCV Distortion and Noise - 400 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	39.97	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_2N
5.2 RCV Distortion and Noise - 500 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	28.09	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_2N
5.2 RCV Distortion and Noise - 630 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	26.67	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_2N
5.2 RCV Distortion and Noise - 800 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	40.18	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_2N
5.2 RCV Distortion and Noise - 1000 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	26.49	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_2N
5.2 RCV Distortion and Noise - 1250 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	26.67	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_2N
5.2 RCV Distortion and Noise - 1600 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	43.16	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_2N
5.2 RCV Distortion and Noise - 2000 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	41.93	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_2N
5.2 RCV Distortion and Noise - 2500 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	41.72	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_2N
5.2 RCV Distortion and Noise - 3150 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	38.20	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_2N
Report - Receive Distortion and Noise (Conversational Gain)	Ok	Minimum SDNR [dB], (occured at 1000Hz)	26.49	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_2N
5.3 Frequency Response 2N FF HANB	Ok	Min. dist. to tolerance scheme [dB], 1360.9 Hz	2.05	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_2N
5.3 Frequency Response 2N DF HANB	Ok	Min. dist. to tolerance scheme [dB], 305.9 Hz	1.20	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_2N

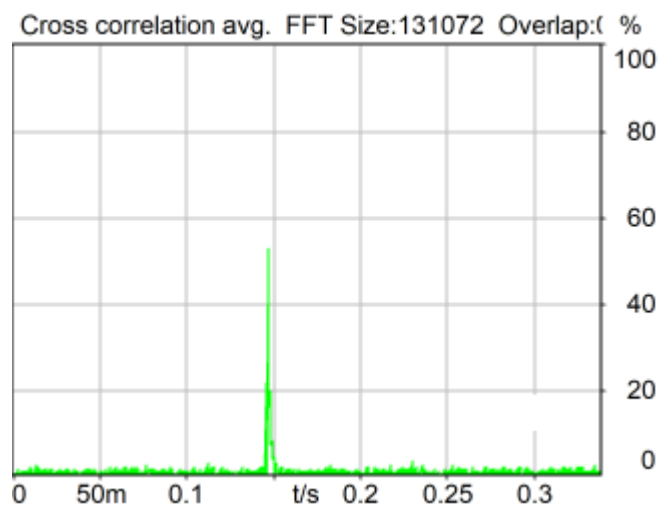
ACQUA Report
25_20M_QPSK_1RB_0Offset_EVS NB 24.4kbps_Ch26340_2N

Title: LTE Band

Overall Receive Delay NB	5
5.1b Receive Volume Control Performance 2N NB	6
5.2 RCV Distortion and Noise - 400 Hz NB	7
5.2 RCV Distortion and Noise - 500 Hz NB	8
5.2 RCV Distortion and Noise - 630 Hz NB	9
5.2 RCV Distortion and Noise - 800 Hz NB	10
5.2 RCV Distortion and Noise - 1000 Hz NB	11
5.2 RCV Distortion and Noise - 1250 Hz NB	12
5.2 RCV Distortion and Noise - 1600 Hz NB	13
5.2 RCV Distortion and Noise - 2000 Hz NB	14
5.2 RCV Distortion and Noise - 2500 Hz NB	15
5.2 RCV Distortion and Noise - 3150 Hz NB	16
Report - Receive Distortion and Noise (Conversational Gain)	17
5.3 Frequency Response 2N FF HANB	18
5.3 Frequency Response 2N DF HANB	19

Overall Receive Delay NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ Preparation - Delay measurement

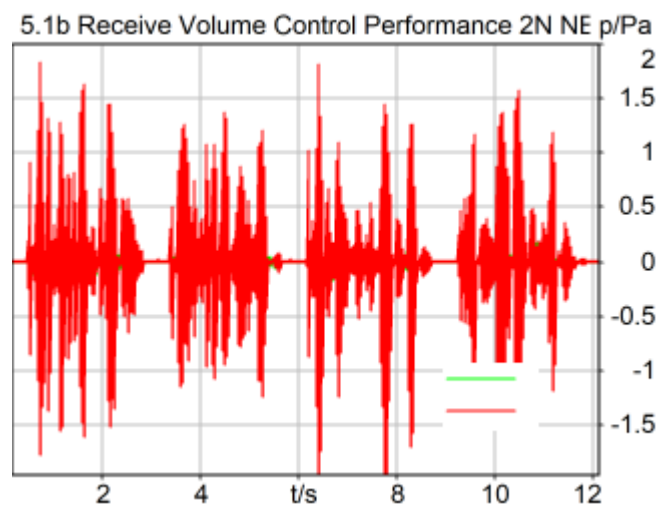


Delay (Cross): 147.4 ms

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5.1b Receive Volume Control Performance 2N NB

TIA-5050 (2018-01) \ Measurements \ Narrowband



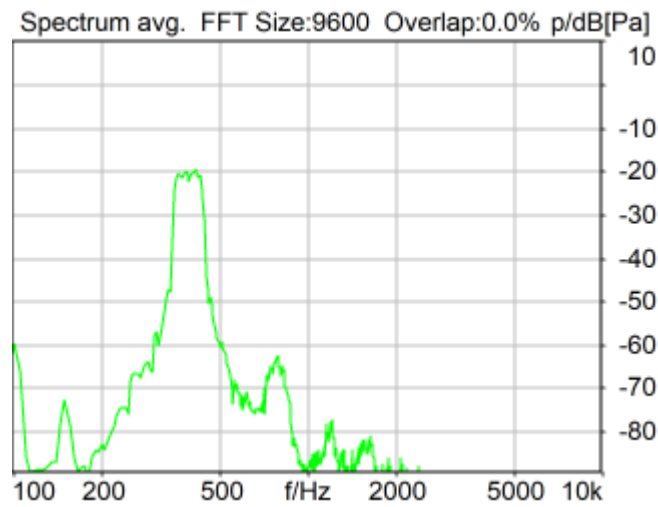
Speech Level RCV: 80.40 dB[SPL], Act.: 80.12% Ok

Ok

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5.2 RCV Distortion and Noise - 400 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



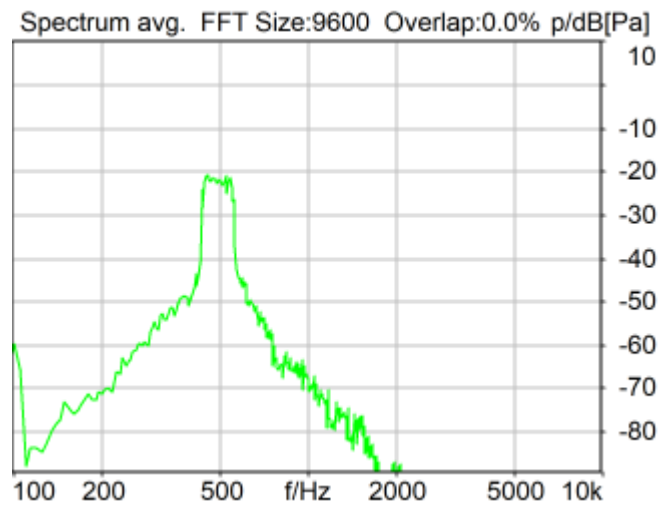
Distortion (Noise) RCV (packed): 39.97 dB (1.00%) Ok

Ok

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5.2 RCV Distortion and Noise - 500 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



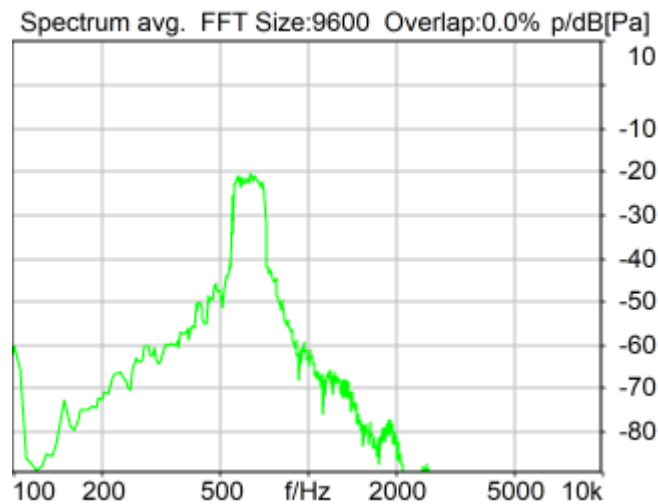
Distortion (Noise) RCV (packed): 28.09 dB (3.94%) Ok

Ok

26/03/13 18:36 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 630 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



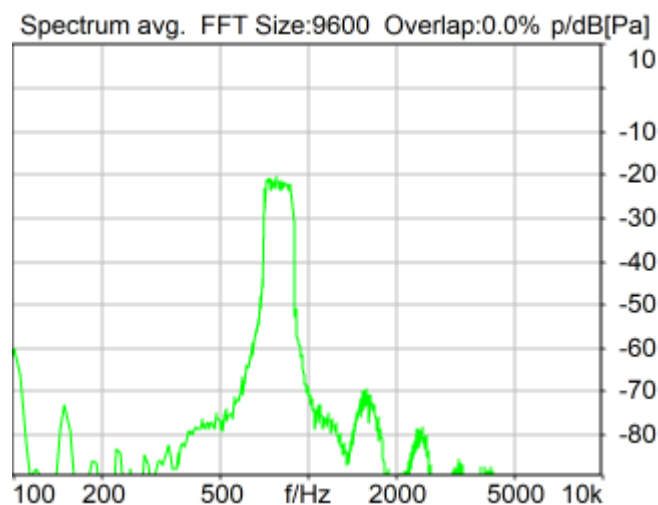
Distortion (Noise) RCV (packed): 26.67 dB (4.64%) Ok

Ok

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5.2 RCV Distortion and Noise - 800 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



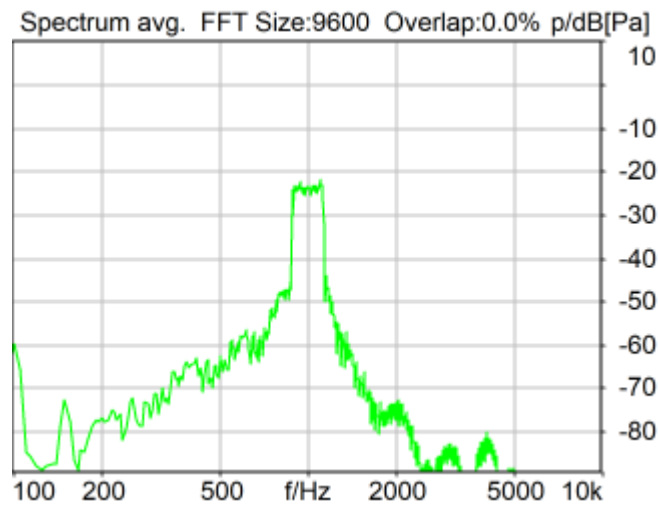
Distortion (Noise) RCV (packed): 40.18 dB (0.98%) Ok

Ok

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5.2 RCV Distortion and Noise - 1000 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



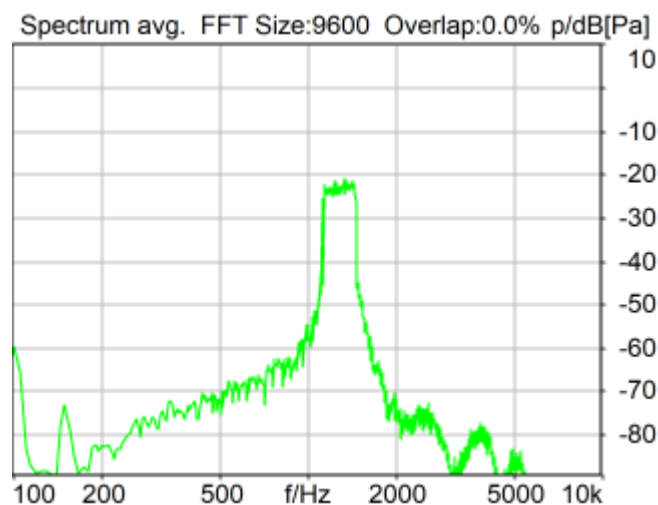
Distortion (Noise) RCV (packed): 26.49 dB (4.74%) Ok

Ok

26/03/13 18:38 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 1250 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



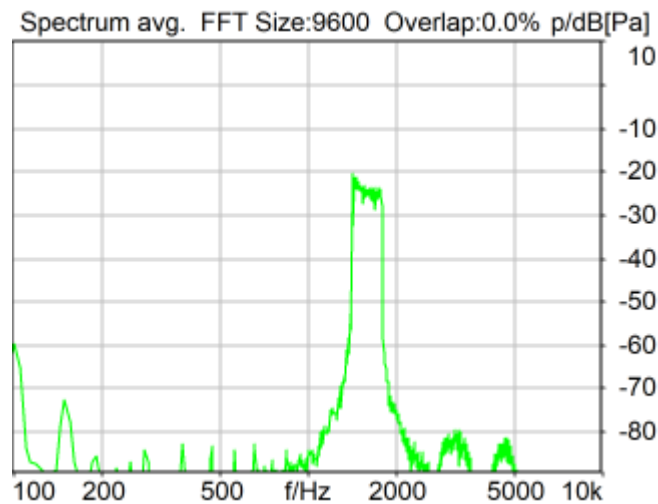
Distortion (Noise) RCV (packed): 26.67 dB (4.64%) Ok

Ok

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5.2 RCV Distortion and Noise - 1600 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



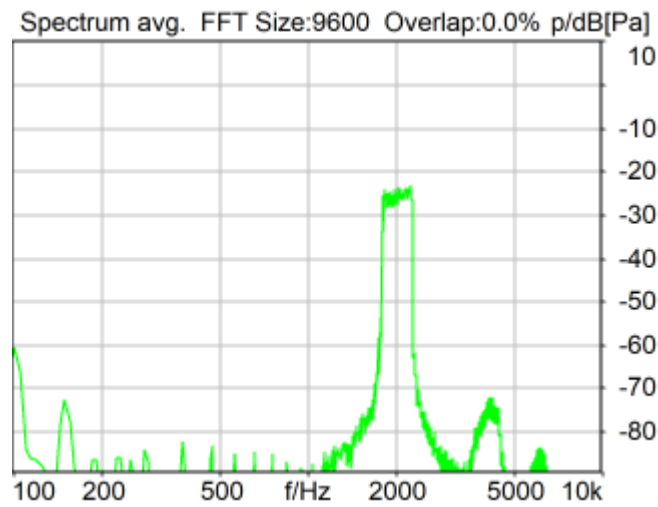
Distortion (Noise) RCV (packed): 43.16 dB (0.69%) Ok

Ok

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5.2 RCV Distortion and Noise - 2000 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



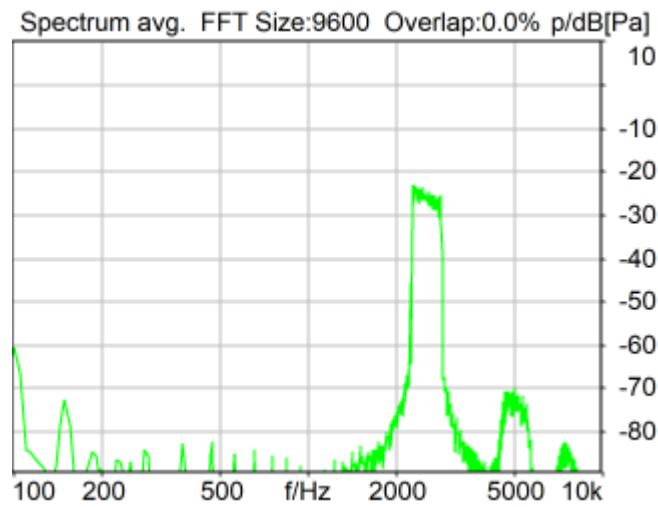
Distortion (Noise) RCV (packed): 41.93 dB (0.80%) Ok

Ok

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5.2 RCV Distortion and Noise - 2500 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



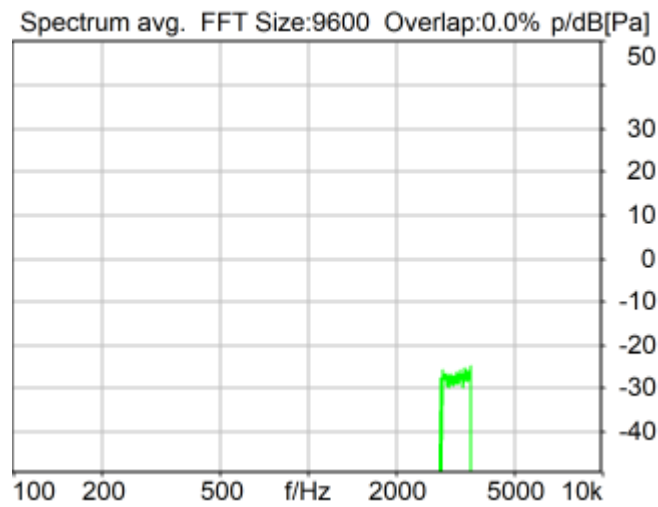
Distortion (Noise) RCV (packed): 41.72 dB (0.82%) Ok

Ok

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5.2 RCV Distortion and Noise - 3150 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



Distortion (Noise) RCV (packed): 38.20 dB (1.23%) Ok

Ok

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Report - Receive Distortion and Noise (Conversational Gain)

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N

Region	Frequency	SDNR
1	400Hz	39.97 dB
2	500Hz	28.09 dB
3	630Hz	26.67 dB
4	800Hz	40.18 dB
5	1000Hz	26.49 dB
6	1250Hz	26.67 dB
7	1600Hz	43.16 dB
8	2000Hz	41.93 dB
9	2500Hz	41.72 dB
10	3150Hz	38.20 dB

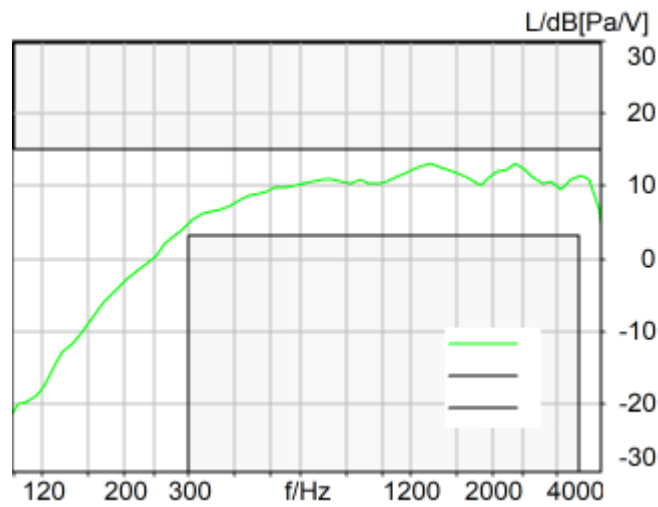
All SDNRs were greater than 20.0 dB, requirement was met.

Smallest SDNR was 26.49dB at 1000Hz.

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5.3 Frequency Response 2N FF HANB

TIA-5050 (2018-01) \ Measurements \ Narrowband



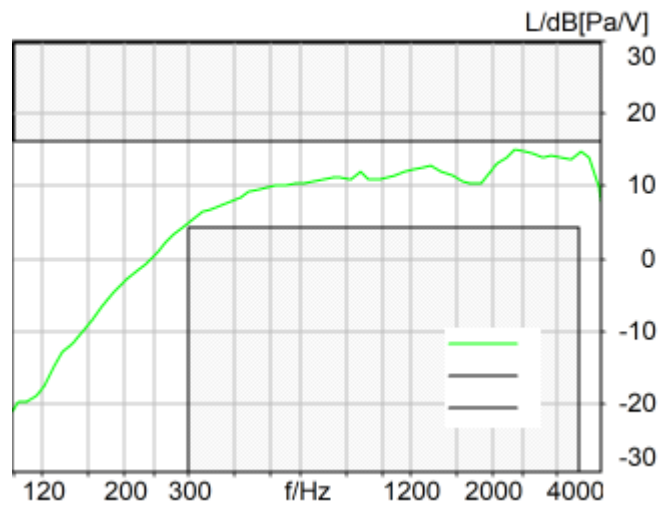
Absolute minimal distance
2.05 dB at 1360.9 Hz Ok

Ok

26/03/13 18:40 ACQUA 5.1.200

5.3 Frequency Response 2N DF HANB

TIA-5050 (2018-01) \ Measurements \ Narrowband



Absolute minimal distance
1.20 dB at 305.9 Hz Ok

Ok

26/03/13 18:41 ACQUA 5.1.200

Measurement Protocol

Measurement Object	LTE Band 25_20M_QPSK_1RB_0Offset_EVS NB 24.4kbps_Ch26340_8N
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Project	612311
Report Generation Date	26/03/13 18:53
Responsible Person	AC01-KS

Status Overview

SMD	Status	Single Value Description	Single Value	Object
Overall Receive Delay NB	Done	Delay (Cross) [ms]	147.9	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_8N
5.1a Receive Volume Control Performance 8N NB	Ok	Speech Level [dB[SPL]]	84.49	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_8N
5.2 RCV Distortion and Noise - 400 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	33.38	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_8N
5.2 RCV Distortion and Noise - 500 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	28.97	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_8N
5.2 RCV Distortion and Noise - 630 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	26.71	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_8N
5.2 RCV Distortion and Noise - 800 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	34.26	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_8N
5.2 RCV Distortion and Noise - 1000 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	41.15	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_8N
5.2 RCV Distortion and Noise - 1250 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	32.83	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_8N
5.2 RCV Distortion and Noise - 1600 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	42.53	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_8N
5.2 RCV Distortion and Noise - 2000 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	44.32	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_8N
5.2 RCV Distortion and Noise - 2500 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	41.93	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_8N
5.2 RCV Distortion and Noise - 3150 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	41.43	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_8N
Report - Receive Distortion and Noise (Conversational Gain)	Ok	Minimum SDNR [dB], (occured at 630Hz)	26.71	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_8N
5.3 Frequency Response 8N FF HANB	Ok	Min. dist. to tolerance scheme [dB], 1360.9 Hz	2.07	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_8N
5.3 Frequency Response 8N DF HANB	Ok	Min. dist. to tolerance scheme [dB], 2302.3 Hz	1.74	LTE Band 25_20M_QPSK_1RB_0Offset_EV S NB 24.4kbps_Ch26340_8N

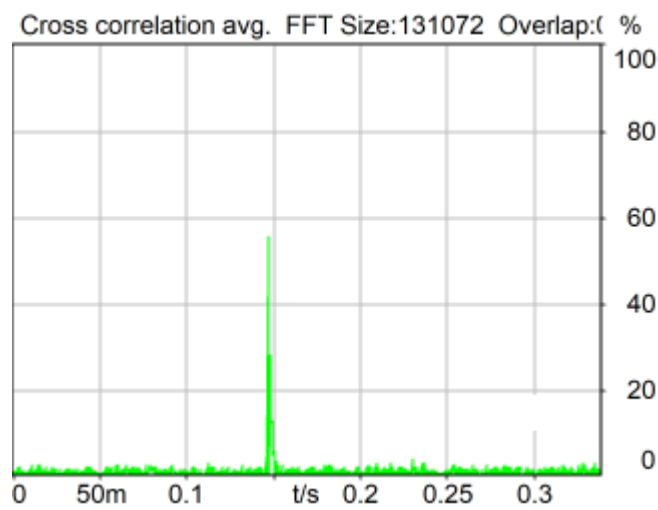
ACQUA Report
25_20M_QPSK_1RB_0Offset_EVS NB 24.4kbps_Ch26340_8N

Title: LTE Band

Overall Receive Delay NB	5
5.1a Receive Volume Control Performance 8N NB	6
5.2 RCV Distortion and Noise - 400 Hz NB	7
5.2 RCV Distortion and Noise - 500 Hz NB	8
5.2 RCV Distortion and Noise - 630 Hz NB	9
5.2 RCV Distortion and Noise - 800 Hz NB	10
5.2 RCV Distortion and Noise - 1000 Hz NB	11
5.2 RCV Distortion and Noise - 1250 Hz NB	12
5.2 RCV Distortion and Noise - 1600 Hz NB	13
5.2 RCV Distortion and Noise - 2000 Hz NB	14
5.2 RCV Distortion and Noise - 2500 Hz NB	15
5.2 RCV Distortion and Noise - 3150 Hz NB	16
Report - Receive Distortion and Noise (Conversational Gain)	17
5.3 Frequency Response 8N FF HANB	18
5.3 Frequency Response 8N DF HANB	19

Overall Receive Delay NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ Preparation - Delay measurement

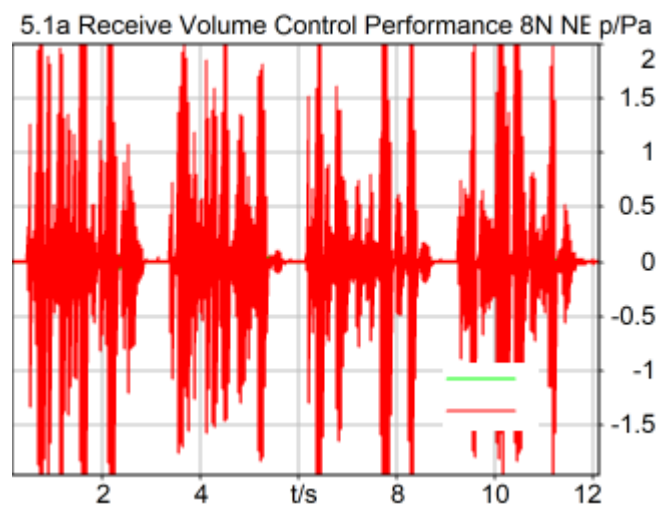


Delay (Cross): 147.9 ms

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5.1a Receive Volume Control Performance 8N NB

TIA-5050 (2018-01) \ Measurements \ Narrowband



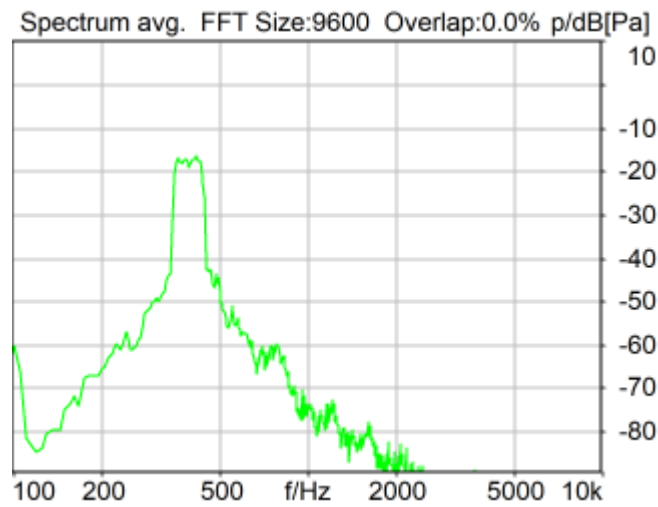
Speech Level RCV: 84.49 dB[SPL], Act.: 80.37% Ok

Ok

26/03/13 18:47 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 400 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 8N



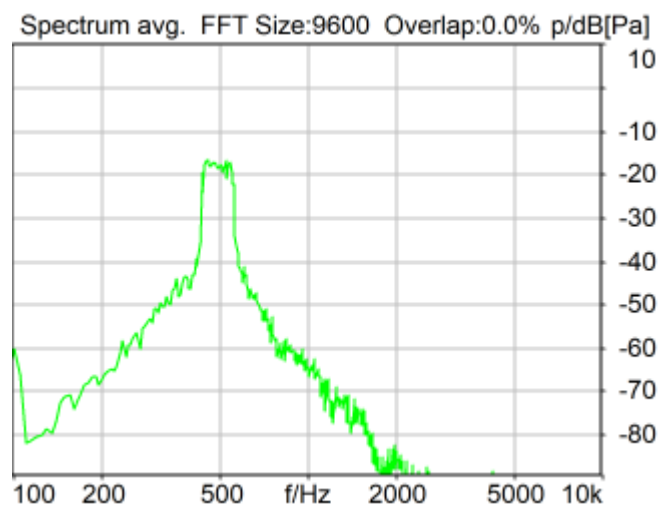
Distortion (Noise) RCV (packed): 33.38 dB (2.14%) Ok

Ok

26/03/13 18:47 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 500 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 8N



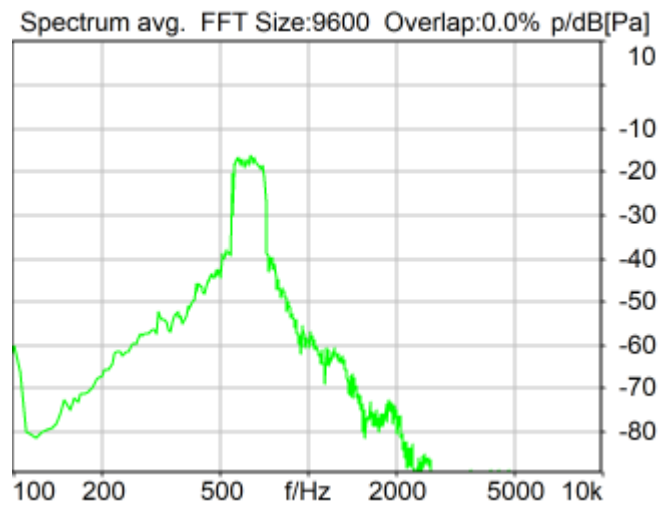
Distortion (Noise) RCV (packed): 28.97 dB (3.56%) Ok

Ok

26/03/13 18:48 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 630 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 8N



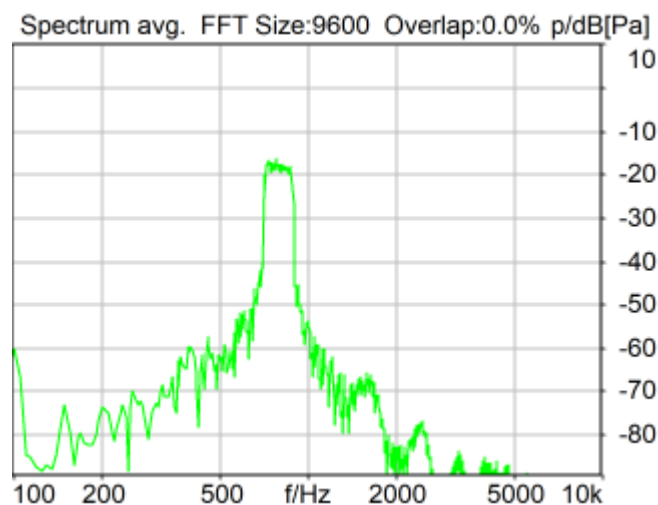
Distortion (Noise) RCV (packed): 26.71 dB (4.62%) Ok

Ok

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5.2 RCV Distortion and Noise - 800 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 8N



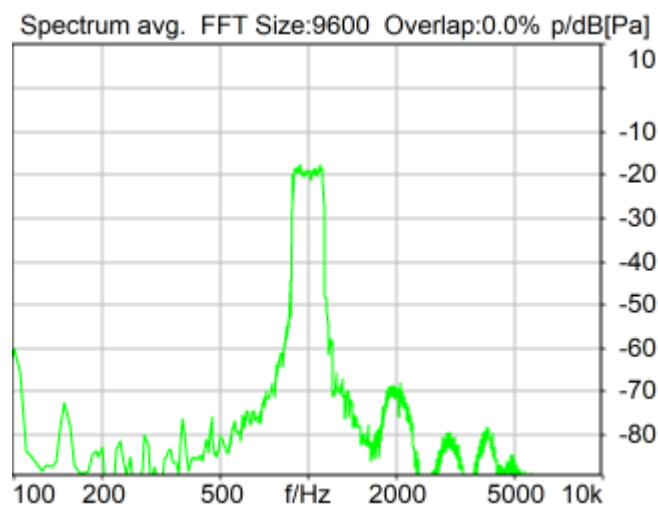
Distortion (Noise) RCV (packed): 34.26 dB (1.94%) Ok

Ok

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5.2 RCV Distortion and Noise - 1000 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 8N



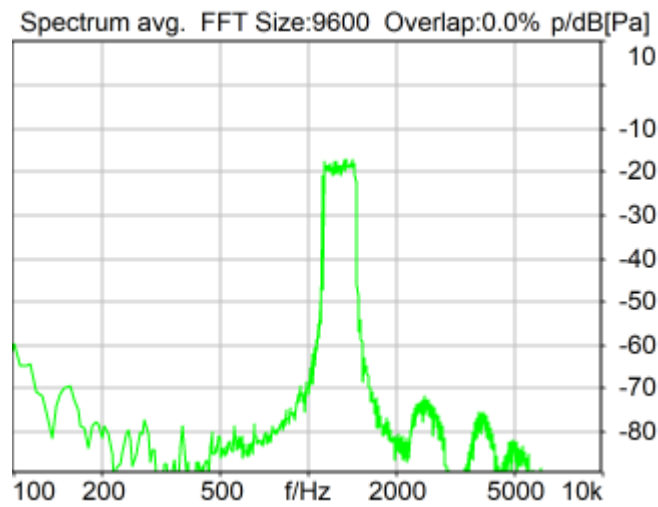
Distortion (Noise) RCV (packed): 41.15 dB (0.88%) Ok

Ok

26/03/13 18:49 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 1250 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 8N



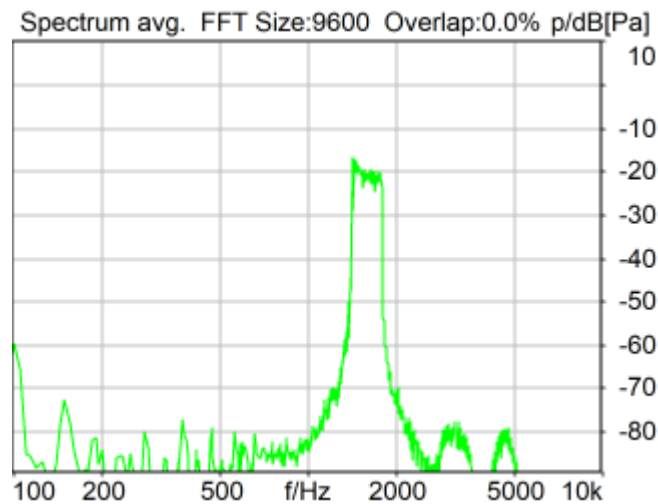
Distortion (Noise) RCV (packed): 32.83 dB (2.28%) Ok

Ok

26/03/13 18:50 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 1600 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 8N



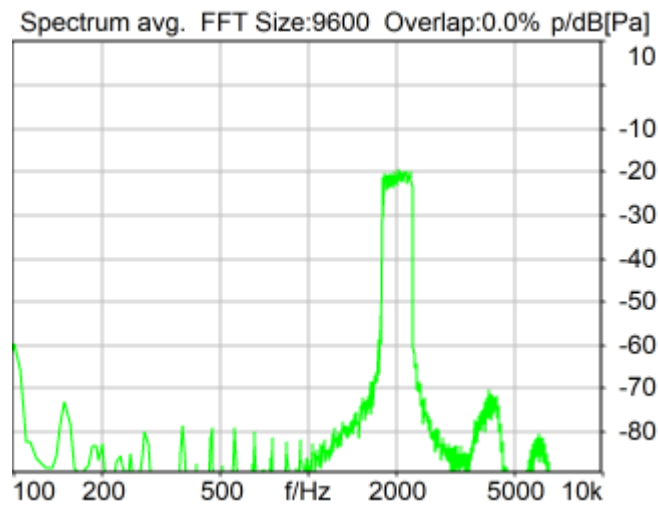
Distortion (Noise) RCV (packed): 42.53 dB (0.75%) Ok

Ok

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5.2 RCV Distortion and Noise - 2000 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 8N



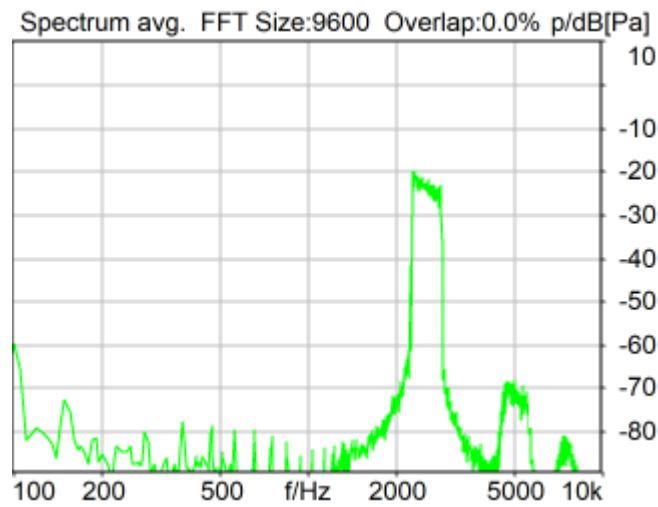
Distortion (Noise) RCV (packed): 44.32 dB (0.61%) Ok

Ok

26/03/13 18:51 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 2500 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 8N



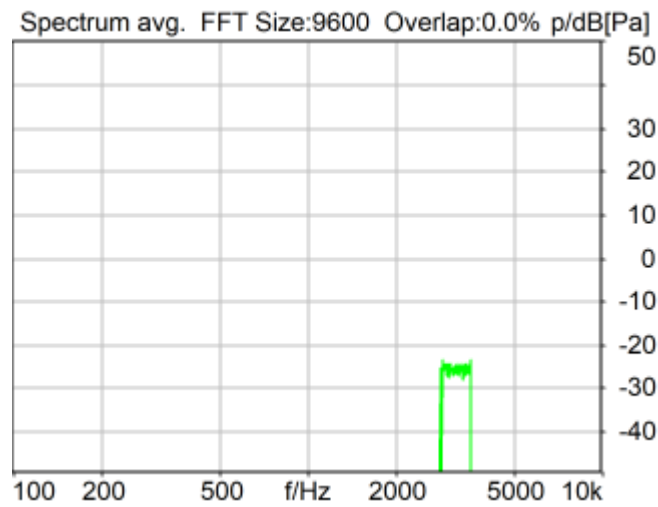
Distortion (Noise) RCV (packed): 41.93 dB (0.80%) Ok

Ok

26/03/13 18:51 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 3150 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 8N



Distortion (Noise) RCV (packed): 41.43 dB (0.85%) Ok

Ok

26/03/13 18:52 ACQUA 5.1.200

Report - Receive Distortion and Noise (Conversational Gain)

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 8N

Region	Frequency	SDNR
1	400Hz	33.38 dB
2	500Hz	28.97 dB
3	630Hz	26.71 dB
4	800Hz	34.26 dB
5	1000Hz	41.15 dB
6	1250Hz	32.83 dB
7	1600Hz	42.53 dB
8	2000Hz	44.32 dB
9	2500Hz	41.93 dB
10	3150Hz	41.43 dB

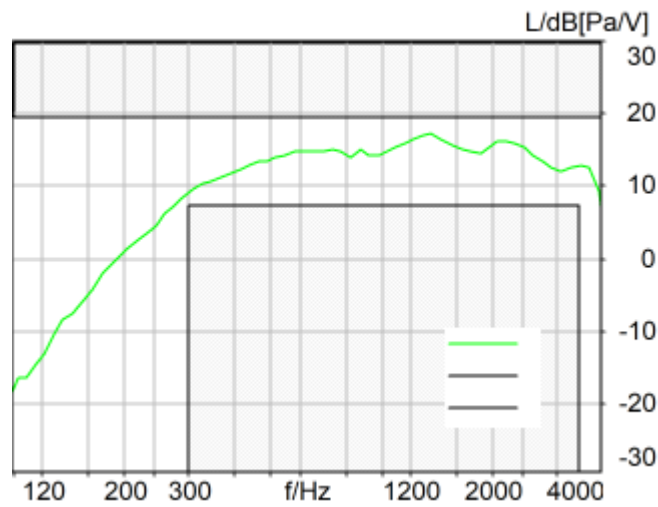
All SDNRs were greater than 20.0 dB, requirement was met.

Smallest SDNR was 26.71dB at 630Hz.

26/03/13 18:52 ACQUA

5.3 Frequency Response 8N FF HANB

TIA-5050 (2018-01) \ Measurements \ Narrowband



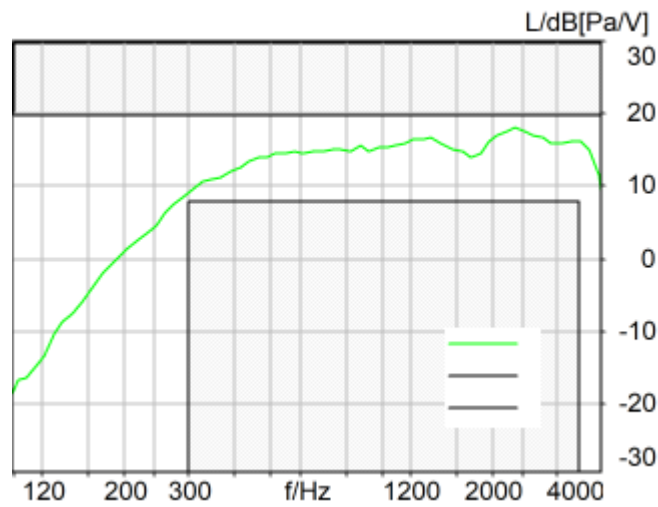
Absolute minimal distance
2.07 dB at 1360.9 Hz Ok

Ok

26/03/13 18:52 ACQUA 5.1.200

5.3 Frequency Response 8N DF HANB

TIA-5050 (2018-01) \ Measurements \ Narrowband



Absolute minimal distance
1.74 dB at 2302.3 Hz Ok

Ok

26/03/13 18:52 ACQUA 5.1.200

Measurement Protocol

Measurement Object	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N
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Project	612311
Report Generation Date	26/04/19 6:53
Responsible Person	AC01-KS

Status Overview

SMD	Status	Single Value Description	Single Value	Object
Overall Receive Delay NB	Done	Delay (Cross) [ms]	274.7	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N
5.1b Receive Volume Control Performance 2N NB	Ok	Speech Level [dB[SPL]]	81.02	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N
5.2 RCV Distortion and Noise - 400 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	37.33	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N
5.2 RCV Distortion and Noise - 500 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	37.42	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N
5.2 RCV Distortion and Noise - 630 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	27.28	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N
5.2 RCV Distortion and Noise - 800 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	30.96	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N
5.2 RCV Distortion and Noise - 1000 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	40.28	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N
5.2 RCV Distortion and Noise - 1250 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	32.03	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N
5.2 RCV Distortion and Noise - 1600 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	29.98	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N
5.2 RCV Distortion and Noise - 2000 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	37.93	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N
5.2 RCV Distortion and Noise - 2500 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	42.57	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N
5.2 RCV Distortion and Noise - 3150 Hz NB	Ok	Distortion (Noise) [dB], 0.0 dB	41.73	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N
Report - Receive Distortion and Noise (Conversational Gain)	Ok	Minimum SDNR [dB], (occured at 630Hz)	27.28	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N
5.3 Frequency Response 2N FF HANB	Ok	Min. dist. to tolerance scheme [dB], 1547.5 Hz	2.11	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N
5.3 Frequency Response 2N DF HANB	Ok	Min. dist. to tolerance scheme [dB], 1547.5 Hz	3.16	FR1 n77_100M_QPSK_1RB_1Offset_EVS NB 24.4kbps_Ch656000_2N

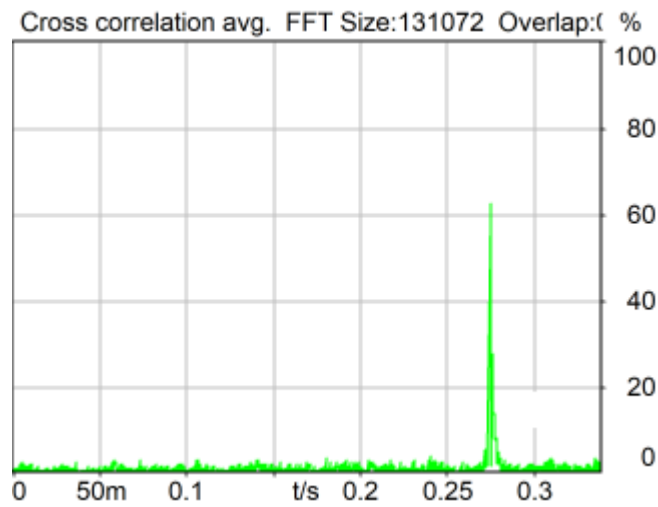
ACQUA Report
NB 24.4kbps_Ch656000_2N

Title: FR1 n77_100M_QPSK_1RB_1Offset_EVS

Overall Receive Delay NB	5
5.1b Receive Volume Control Performance 2N NB	6
5.2 RCV Distortion and Noise - 400 Hz NB	7
5.2 RCV Distortion and Noise - 500 Hz NB	8
5.2 RCV Distortion and Noise - 630 Hz NB	9
5.2 RCV Distortion and Noise - 800 Hz NB	10
5.2 RCV Distortion and Noise - 1000 Hz NB	11
5.2 RCV Distortion and Noise - 1250 Hz NB	12
5.2 RCV Distortion and Noise - 1600 Hz NB	13
5.2 RCV Distortion and Noise - 2000 Hz NB	14
5.2 RCV Distortion and Noise - 2500 Hz NB	15
5.2 RCV Distortion and Noise - 3150 Hz NB	16
Report - Receive Distortion and Noise (Conversational Gain)	17
5.3 Frequency Response 2N FF HANB	18
5.3 Frequency Response 2N DF HANB	19

Overall Receive Delay NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ Preparation - Delay measurement

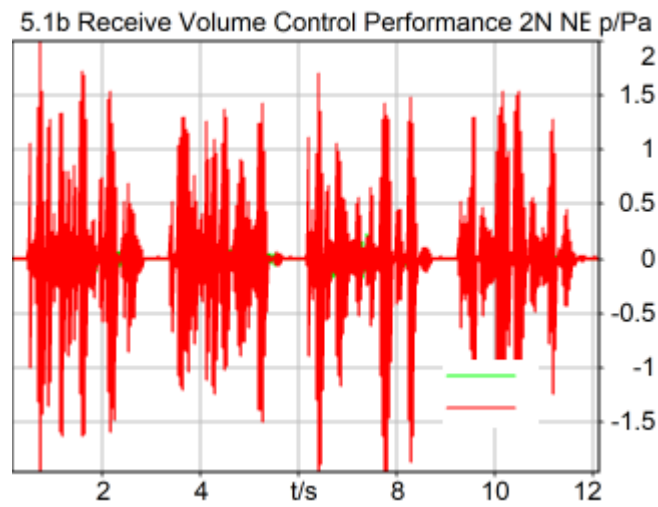


Delay (Cross): 274.7 ms

26/04/19 6:23 ACQUA 5.1.200

5.1b Receive Volume Control Performance 2N NB

TIA-5050 (2018-01) \ Measurements \ Narrowband



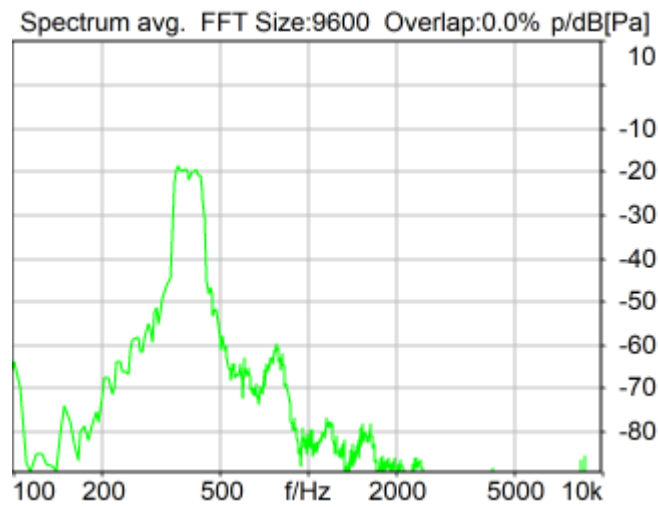
Speech Level RCV: 81.02 dB[SPL], Act.: 80.29% Ok

Ok

26/04/19 6:24 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 400 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



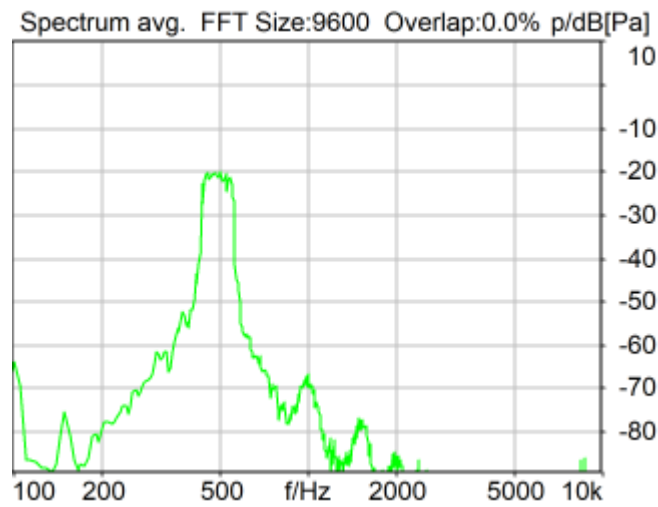
Distortion (Noise) RCV (packed): 37.33 dB (1.36%) Ok

Ok

26/04/19 6:25 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 500 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



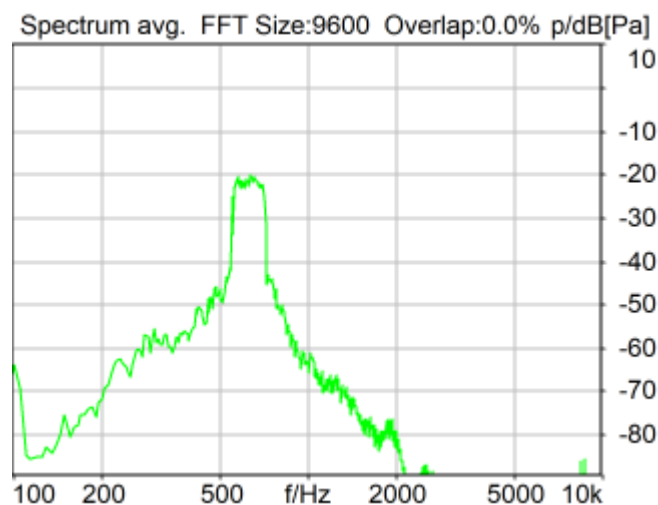
Distortion (Noise) RCV (packed): 37.42 dB (1.35%) Ok

Ok

26/04/19 6:25 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 630 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



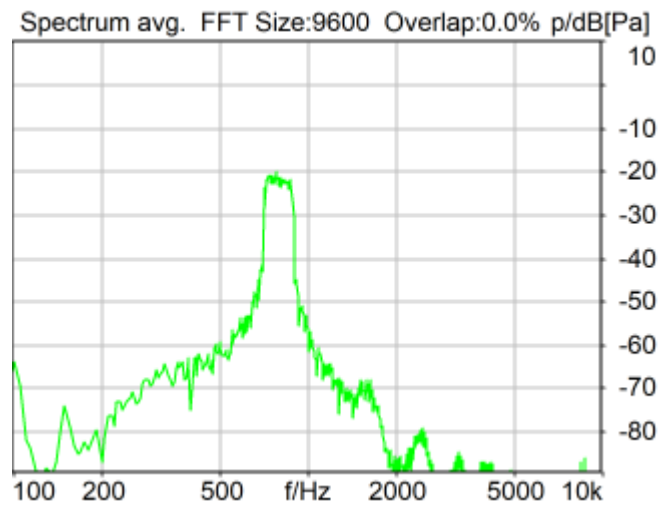
Distortion (Noise) RCV (packed): 27.28 dB (4.33%) Ok

Ok

26/04/19 6:26 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 800 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



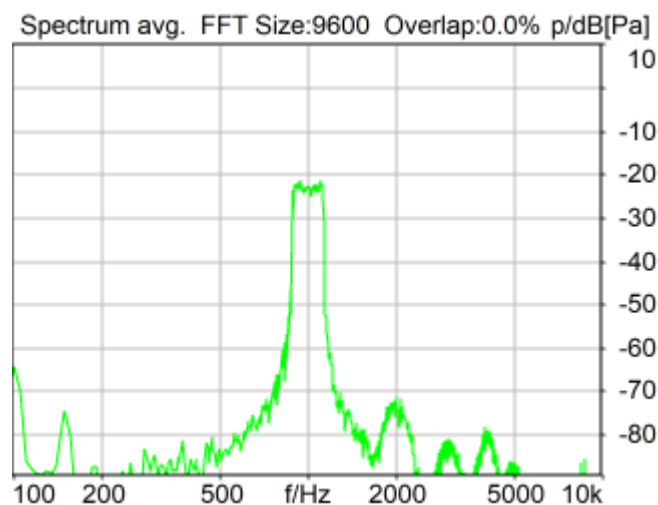
Distortion (Noise) RCV (packed): 30.96 dB (2.83%) Ok

Ok

26/04/19 6:26 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 1000 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



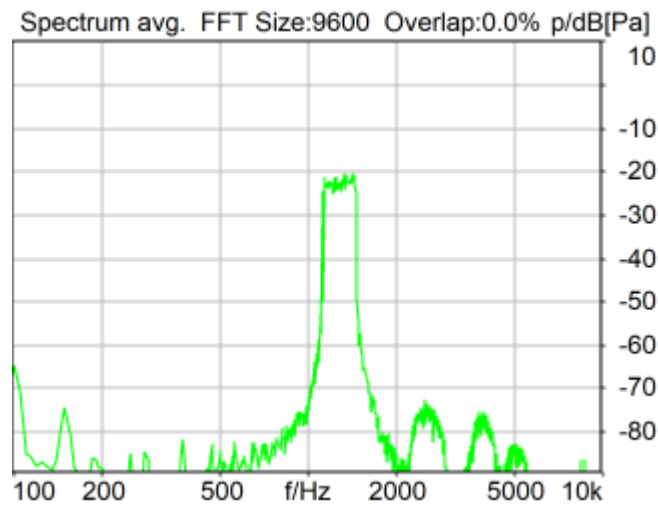
Distortion (Noise) RCV (packed): 40.28 dB (0.97%) Ok

Ok

26/04/19 6:27 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 1250 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



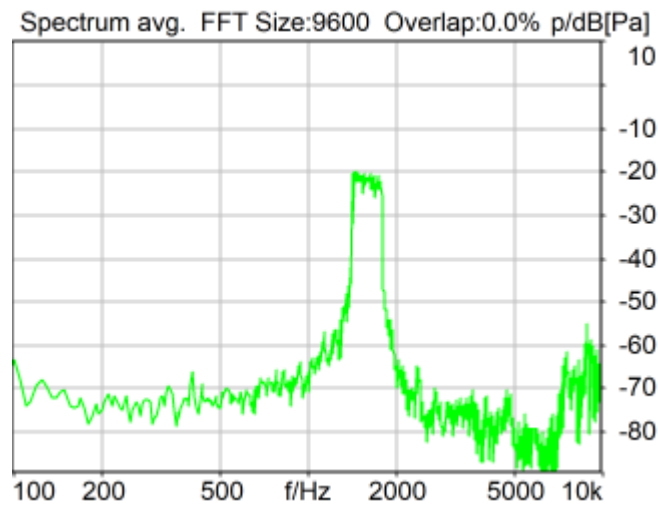
Distortion (Noise) RCV (packed): 32.03 dB (2.50%) Ok

Ok

26/04/19 6:27 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 1600 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



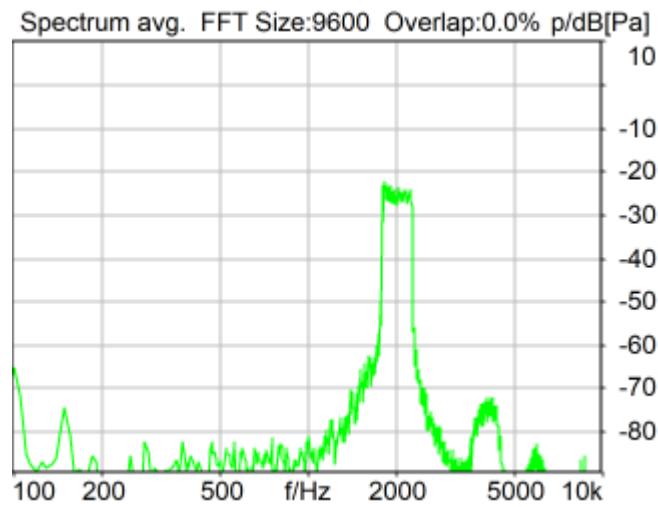
Distortion (Noise) RCV (packed): 29.98 dB (3.17%) Ok

Ok

26/04/19 6:28 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 2000 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



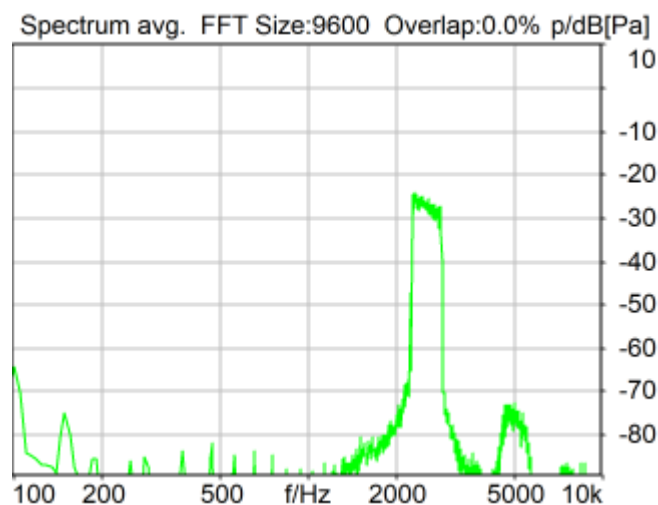
Distortion (Noise) RCV (packed): 37.93 dB (1.27%) Ok

Ok

26/04/19 6:28 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 2500 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



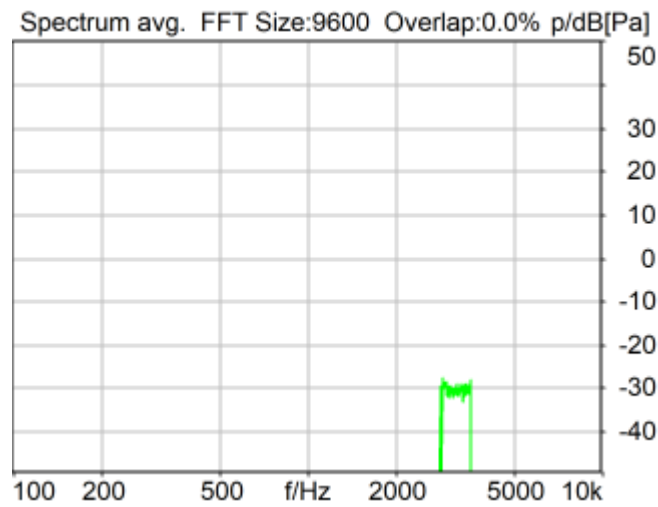
Distortion (Noise) RCV (packed): 42.57 dB (0.74%) Ok

Ok

26/04/19 6:29 ACQUA 5.1.200

5.2 RCV Distortion and Noise - 3150 Hz NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N



Distortion (Noise) RCV (packed): 41.73 dB (0.82%) Ok

Ok

26/04/19 6:29 ACQUA 5.1.200

Report - Receive Distortion and Noise (Conversational Gain)

TIA-5050 (2018-01) \ Measurements \ Narrowband \ 5.2 Receive Distortion and Noise 2N

Region	Frequency	SDNR
1	400Hz	37.33 dB
2	500Hz	37.42 dB
3	630Hz	27.28 dB
4	800Hz	30.96 dB
5	1000Hz	40.28 dB
6	1250Hz	32.03 dB
7	1600Hz	29.98 dB
8	2000Hz	37.93 dB
9	2500Hz	42.57 dB
10	3150Hz	41.73 dB

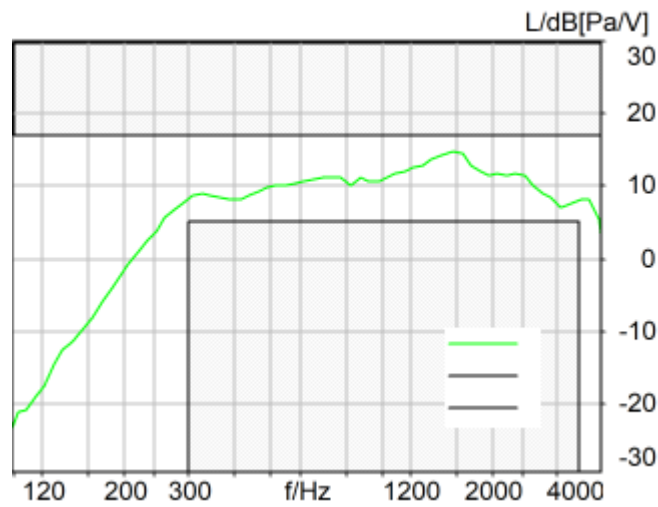
All SDNRs were greater than 20.0 dB, requirement was met.

Smallest SDNR was 27.28dB at 630Hz.

26/04/19 6:52 ACQUA

5.3 Frequency Response 2N FF HANB

TIA-5050 (2018-01) \ Measurements \ Narrowband



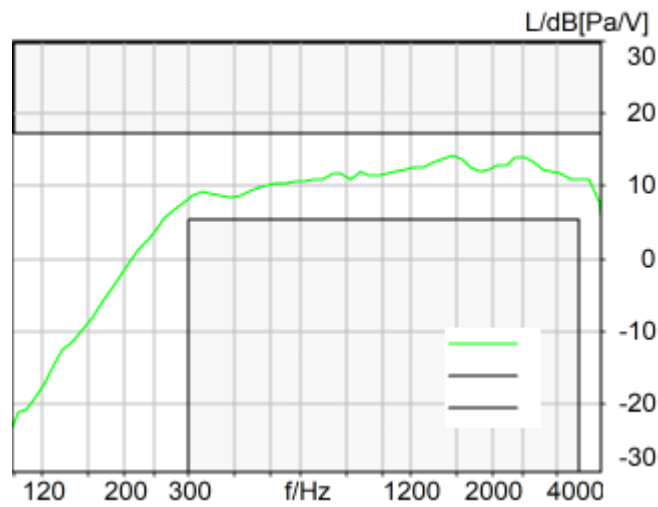
Absolute minimal distance
2.11 dB at 1547.5 Hz Ok

Ok

26/04/19 6:52 ACQUA 5.1.200

5.3 Frequency Response 2N DF HANB

TIA-5050 (2018-01) \ Measurements \ Narrowband



Absolute minimal distance
3.16 dB at 1547.5 Hz Ok

Ok

26/04/19 6:53 ACQUA 5.1.200

Measurement Protocol

Measurement Object	GSM850_GSM Voice_AMR NB 4.75kbps_Ch189_2N
--------------------	---

Project	612311
Report Generation Date	26/03/20 14:04
Responsible Person	AC01-KS

Status Overview

SMD	Status	Single Value Description	Single Value	Object
Overall Receive Delay NB	Done	Delay (Cross) [ms]	198.2	GSM850_GSM Voice_AMR NB 4. 75kbps_Ch189_2N
5.1b Receive Volume Control Performance 2N NB	Ok	Speech Level [dB[SPL]]	80.34	GSM850_GSM Voice_AMR NB 4. 75kbps_Ch189_2N

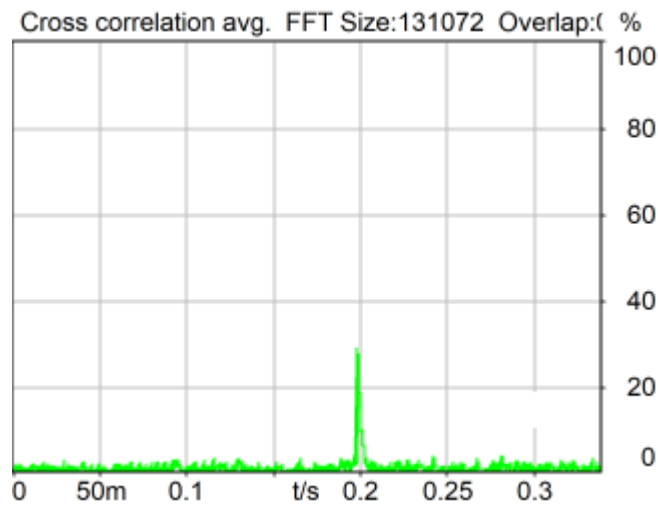
ACQUA Report
75kbps_Ch189_2N

Title: GSM850_GSM Voice_AMR NB 4.

Overall Receive Delay NB	5
5.1b Receive Volume Control Performance 2N NB	6

Overall Receive Delay NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ Preparation - Delay measurement

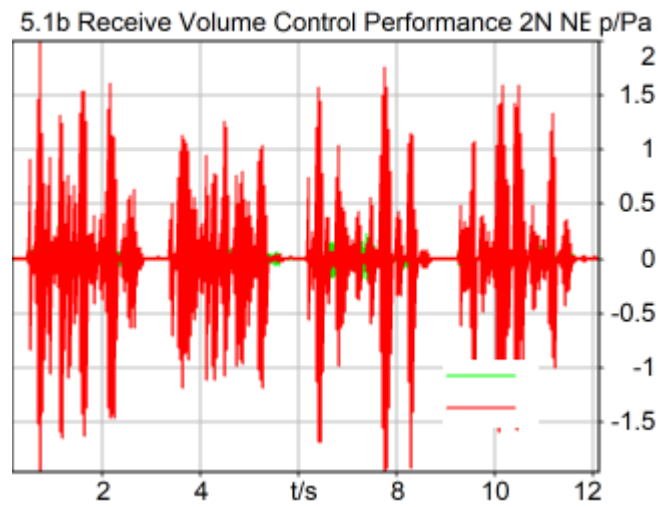


Delay (Cross): 198.2 ms

26/03/20 14:03 ACQUA 5.1.200

5.1b Receive Volume Control Performance 2N NB

TIA-5050 (2018-01) \ Measurements \ Narrowband



Speech Level RCV: 80.34 dB[SPL], Act.: 79.82% Ok

Ok

26/03/20 14:03 ACQUA 5.1.200

Measurement Protocol

Measurement Object	WCDMA V_Voice_AMR NB 4.75kbps_Ch4182_2N
--------------------	---

Project	612311
Report Generation Date	26/03/20 15:24
Responsible Person	AC01-KS

Status Overview

SMD	Status	Single Value Description	Single Value	Object
Overall Receive Delay NB	Done	Delay (Cross) [ms]	216.6	WCDMA V_Voice_AMR NB 4. 75kbps_Ch4182_2N
5.1b Receive Volume Control Performance 2N NB	Ok	Speech Level [dB[SPL]]	80.27	WCDMA V_Voice_AMR NB 4. 75kbps_Ch4182_2N

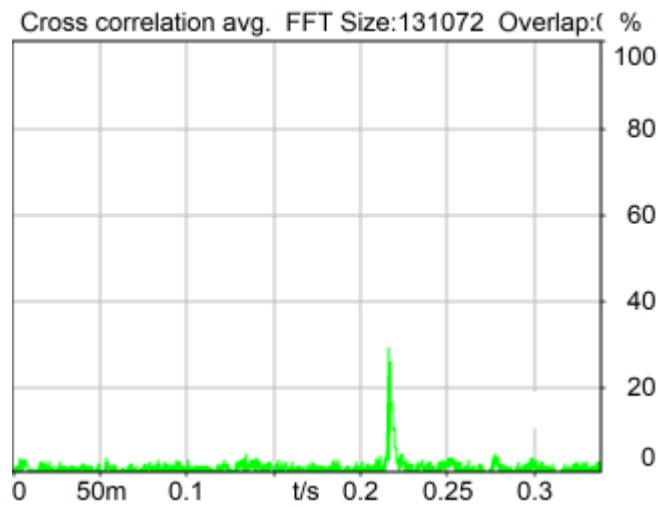
ACQUA Report
75kbps_Ch4182_2N

Title: WCDMA V_Voice_AMR NB 4.

Overall Receive Delay NB	5
5.1b Receive Volume Control Performance 2N NB	6

Overall Receive Delay NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ Preparation - Delay measurement

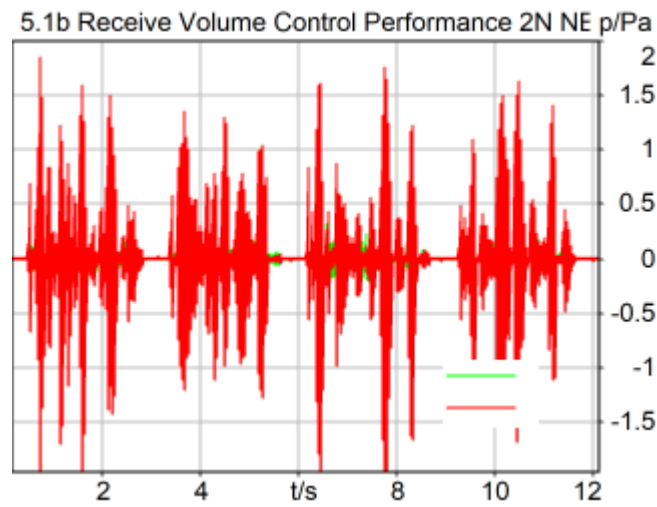


Delay (Cross): 216.6 ms

26/03/20 15:23 ACQUA 5.1.200

5.1b Receive Volume Control Performance 2N NB

TIA-5050 (2018-01) \ Measurements \ Narrowband



Speech Level RCV: 80.27 dB[SPL], Act.: 79.82% Ok

Ok

26/03/20 15:23 ACQUA 5.1.200

Measurement Protocol

Measurement Object	LTE Band 25_20M_QPSK_1RB_0Offset_AMR NB 4.75kbps_Ch26340_8N
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Project	612311
Report Generation Date	26/03/20 10:56
Responsible Person	AC01-KS

Status Overview

SMD	Status	Single Value Description	Single Value	Object
Overall Receive Delay NB	Done	Delay (Cross) [ms]	149.7	LTE Band 25_20M_QPSK_1RB_0Offset_AMR NB 4.75kbps_Ch26340_8N
5.1a Receive Volume Control Performance 8N NB	Ok	Speech Level [dB[SPL]]	84.16	LTE Band 25_20M_QPSK_1RB_0Offset_AMR NB 4.75kbps_Ch26340_8N

ACQUA Report

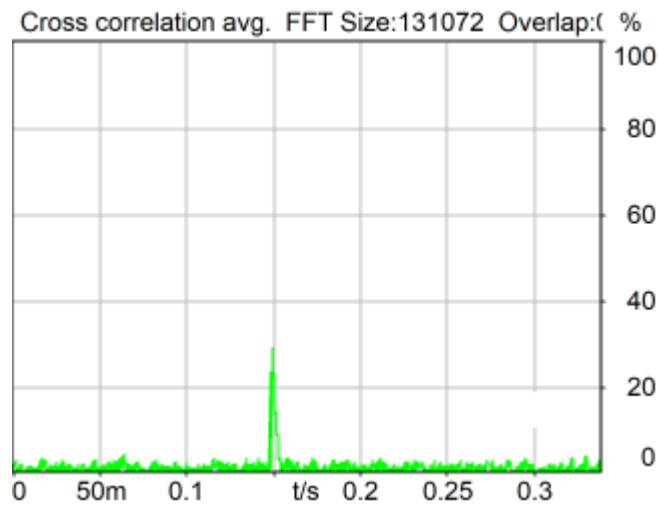
Title: LTE Band

25_20M_QPSK_1RB_0Offset_AMR NB 4.75kbps_Ch26340_8N

Overall Receive Delay NB	5
5.1a Receive Volume Control Performance 8N NB	6

Overall Receive Delay NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ Preparation - Delay measurement

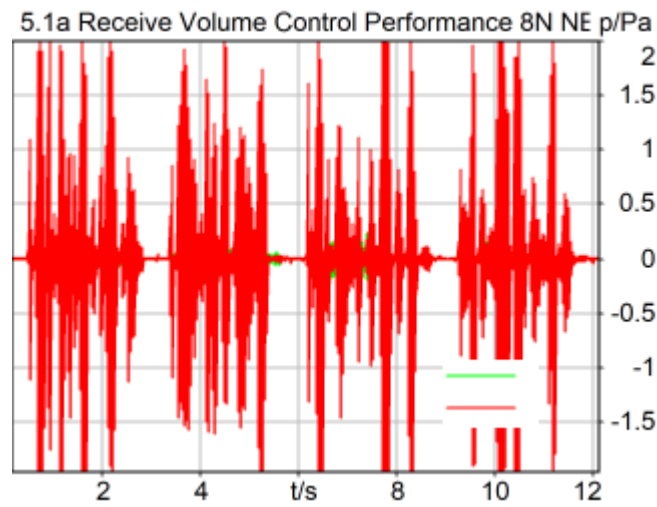


Delay (Cross): 149.7 ms

26/03/20 10:56 ACQUA 5.1.200

5.1a Receive Volume Control Performance 8N NB

TIA-5050 (2018-01) \ Measurements \ Narrowband



Speech Level RCV: 84.16 dB[SPL], Act.: 80.20% Ok

Ok

26/03/20 10:56 ACQUA 5.1.200

Measurement Protocol

Measurement Object	LTE Band 42_20M_QPSK_1RB_0Offset_AMR NB 4.75kbps_Ch42590_2N
--------------------	---

Project	612311
Report Generation Date	26/03/18 16:31
Responsible Person	AC01-KS

Status Overview

SMD	Status	Single Value Description	Single Value	Object
Overall Receive Delay NB	Done	Delay (Cross) [ms]	261.3	LTE Band 42_20M_QPSK_1RB_0Offset_AMR NB 4.75kbps_Ch42590_2N
5.1b Receive Volume Control Performance 2N NB	Ok	Speech Level [dB[SPL]]	79.97	LTE Band 42_20M_QPSK_1RB_0Offset_AMR NB 4.75kbps_Ch42590_2N

ACQUA Report

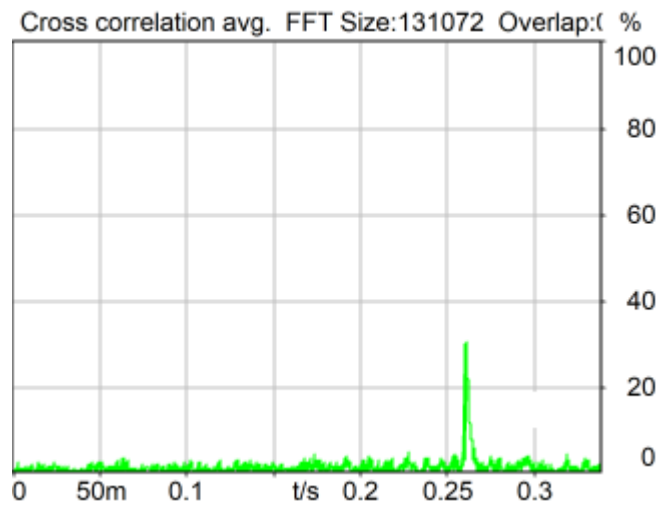
Title: LTE Band

42_20M_QPSK_1RB_0Offset_AMR NB 4.75kbps_Ch42590_2N

Overall Receive Delay NB	5
5.1b Receive Volume Control Performance 2N NB	6

Overall Receive Delay NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ Preparation - Delay measurement

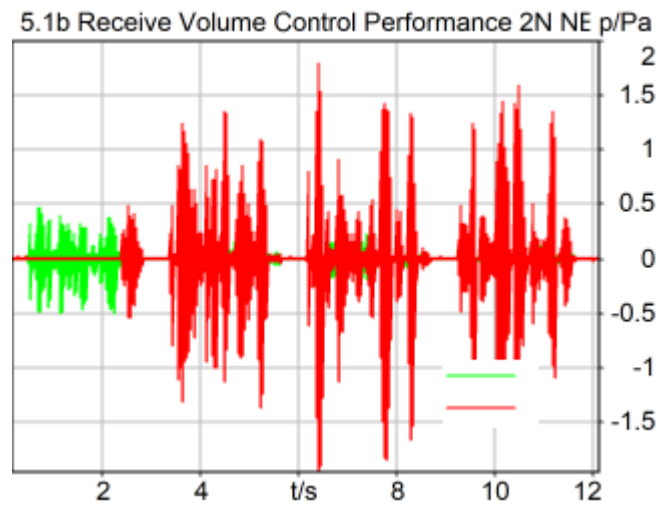


Delay (Cross): 261.3 ms

26/03/18 16:28 ACQUA 5.1.200

5.1b Receive Volume Control Performance 2N NB

TIA-5050 (2018-01) \ Measurements \ Narrowband



Speech Level RCV: 79.97 dB[SPL], Act.: 63.17% Ok

Ok

26/03/18 16:28 ACQUA 5.1.200

Measurement Protocol

Measurement Object	FR1 n5_20M_QPSK_1RB_1Offset_EVS NB 5.9kbps_Ch167300_2N
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Project	612311
Report Generation Date	26/04/20 14:10
Responsible Person	AC01-KS

Status Overview

SMD	Status	Single Value Description	Single Value	Object
Overall Receive Delay NB	Done	Delay (Cross) [ms]	268.5	FR1 n5_20M_QPSK_1RB_1Offset_EVS NB 5.9kbps_Ch167300_2N
5.1b Receive Volume Control Performance 2N NB	Ok	Speech Level [dB[SPL]]	80.36	FR1 n5_20M_QPSK_1RB_1Offset_EVS NB 5.9kbps_Ch167300_2N

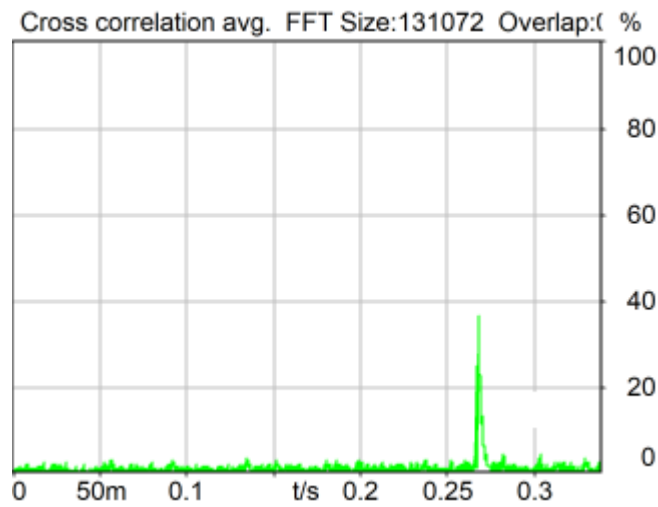
ACQUA Report
5.9kbps_Ch167300_2N

Title: FR1 n5_20M_QPSK_1RB_1Offset_EVS NB

Overall Receive Delay NB	5
5.1b Receive Volume Control Performance 2N NB	6

Overall Receive Delay NB

TIA-5050 (2018-01) \ Measurements \ Narrowband \ Preparation - Delay measurement

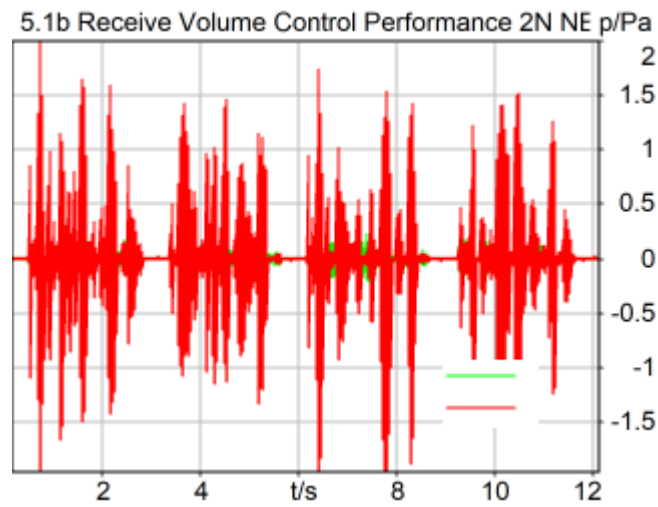


Delay (Cross): 268.5 ms

26/04/20 14:09 ACQUA 5.1.200

5.1b Receive Volume Control Performance 2N NB

TIA-5050 (2018-01) \ Measurements \ Narrowband



Speech Level RCV: 80.36 dB[SPL], Act.: 80.27% Ok

Ok

26/04/20 14:10 ACQUA 5.1.200